Fine-Tuning & LoRA

Module 8 · Zero to LLM

Three ways to adapt

	trains	cost	risk
prompting	nothing	~0	low
LoRA	<1 %	low	low
full fine-tune	100 %	high	forgetting

Instruction format

```
### Instruction: Explain attention in one line with an analogy.
### Response: Attention is a group chat where…<eos>
```

Loss **only** on the response (labels = −100 elsewhere)

LoRA = sticky notes

`y = W x + (α/r) · B A x` · W frozen · A: d×r · B: r×d, **starts at 0**

768×2304 ≈ 1.77 M → r = 8 adds 24.6 k per layer

Failure modes

memorising · forgetting (LR too high) · messy data · no held-out set

Next: M09 — prompts & apps